IMAGE SENSOR ENGINEERING

CCD vs CMOS Sensors

A Technical Comparison of Architecture, Performance, and Engineering Trade-offs

CCD

CMOS

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AGENDA

What We'll Cover

01
Sensor Fundamentals
How photons become a digital image in both technologies

02
Architecture & Signal Path
Charge transfer vs. active pixel readout

03
Technical Parameter Comparison
Noise, dynamic range, speed, power, cost

04
Performance Benchmarks
Quantified trade-offs across key metrics

05
Engineering Trade-offs & Applications
Where each sensor architecture wins

06
Selection Guidance
Matching sensor type to system requirements

A Shared Starting Point, A Different Path

Both technologies use silicon photodiodes to convert incoming photons into electrical charge. They diverge entirely in how that charge is measured and read out.

Photon Capture → Charge Generation → Charge → Voltage Conversion → Amplification & Readout → Digitization (ADC)

Where the divergence begins →

CCD — Charge-Coupled Device

- Charge is physically shifted, pixel by pixel, through the silicon substrate to a single output node.
- Only one (or a few) high-precision amplifier converts charge to voltage for the entire array.
- Analogous to a "bucket brigade" passing charge packets down a line.

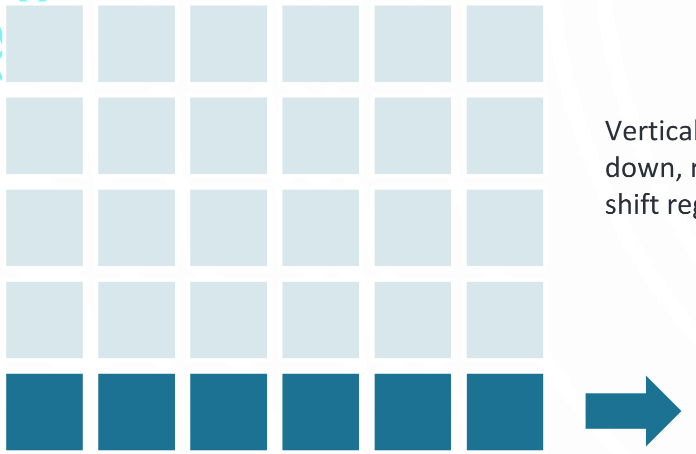
CMOS — Active Pixel Sensor

- Each pixel integrates its own amplifier (and often ADC), converting charge to voltage on-site.
- Pixels are addressed like memory cells — rows and columns selected electronically.
- Enables massively parallel, high-speed readout across the array.

ARCHITECTURE — CCD

CCD Signal Path: Sequential Charge Transfer

Photodiode array



Vertical shift registers move charge down, row by row, into the horizontal shift register (bottom row).

Horizontal Shift Register → **Single Output Amplifier** → **Correlated Double Sampling** → **ADC**

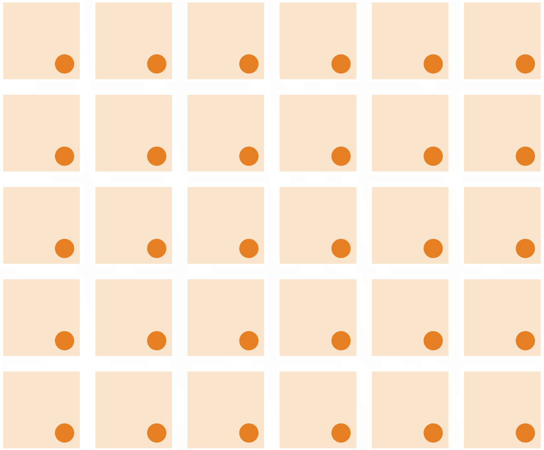
Engineering consequence

- Every pixel is read through the same amplifier → excellent pixel-to-pixel uniformity and low fixed-pattern noise.
- Serial transfer of the entire frame → slower readout and higher clocking power (multiple drive voltages, ~15–30V).

ARCHITECTURE — CMOS

CMOS Signal Path: Parallel Pixel-Level Readout

Each pixel = photodiode + amplifier



Row and column decoders address pixels electronically — no physical charge transfer between pixels.

Row / Column Decoders → **Per-Column ADC Bank** → **Digital Signal Processing** → **Digital Output**

Engineering consequence

- Parallel, column-level ADCs → dramatically faster frame rates and much lower power (single low-voltage supply).
- Per-pixel amplifier variation → higher fixed-pattern noise, mitigated by correlated double sampling (CDS).

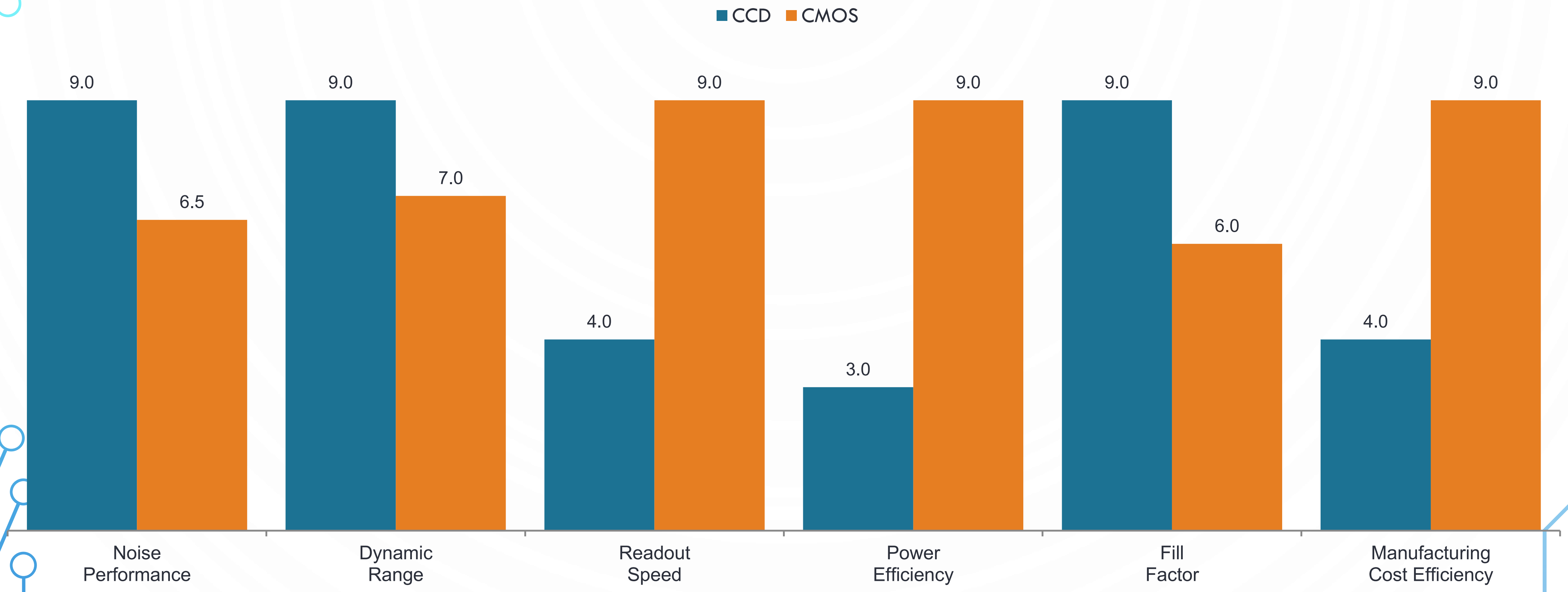
TECHNICAL COMPARISON

Key Parameters, Side by Side

Parameter	CCD	CMOS
Readout architecture	Serial, single output amplifier	Parallel, per-pixel / per-column
Typical readout speed	Lower (tens of fps, frame-limited)	High (hundreds of fps achievable)
Power consumption	High (multiple clock voltages)	Low (single supply, ~1/3–1/10 of CCD)
Read / dark noise	Very low, highly uniform	Higher, improving with CDS designs
Dynamic range	High, typically >70 dB	Moderate–high, 60–75 dB (design-dependent)
Fill factor	Near 100% (front-illuminated)	Lower natively; recovered via microlenses
Blooming / smear	Susceptible to charge overflow	Minimal — pixels reset independently
Shutter type	Native global shutter	Rolling shutter common; global shutter costs area
Fabrication	Specialized, low-volume process	Standard CMOS fabs — high volume, low cost
On-chip integration	Sensor only; external support ICs	ADC, timing, and logic integrated (SoC)

Relative Performance Across Key Metrics

Illustrative index scores (0–10, higher is better) reflecting typical, design-dependent behavior — not fixed physical constants.



Where Each Architecture Wins

CCD — Precision over Speed

Best fit when:

- Absolute image quality and uniformity outweigh speed or power budget
- Very low light levels demand minimal noise (astronomy, spectroscopy)
- Scientific / metrology imaging needs precise, repeatable pixel response
- High-end machine vision and some medical imaging (X-ray, microscopy)

CMOS — Speed, Power & Integration

Best fit when:

- High frame rates, low power, and compact system integration are priorities
- Battery-powered or space-constrained devices (smartphones, wearables)
- Real-time, high-speed capture is required (automotive, robotics, sports)
- Cost-sensitive, high-volume production (consumer electronics, security)

The Engineering Verdict

Neither architecture is universally superior — the choice is a systems-engineering trade-off between image fidelity and speed / power / cost.

CCD

Choose for uncompromising image quality where power and speed are secondary.

- Superior noise & uniformity
- Native global shutter
- Higher power & cost, slower readout

CMOS

Choose for speed, power efficiency, integration, and cost at scale.

- High speed, low power, on-chip SoC
- Low-cost, high-volume fabrication
- Historically higher noise; now closing the gap


Thank You!